

3-MONTH GIT & JENKINS PIPELINE MASTER PLAN

Month 1 — Git, GitHub & DevOps Basics (Weeks 1–4)

Week 1 — Git Basics

- Git fundamentals, clone/pull/push/commit
- Branching & merging
- Practice: repos, branches, PRs

Week 2 — GitHub Collaboration

- Pull requests, reviews
- Branch protection rules
- Conflict resolution

Week 3 — DevOps + Jenkins Intro

- What is CI/CD
- Jenkins architecture
- Freestyle jobs

Week 4 — Jenkins Pipeline Fundamentals

- Declarative vs Scripted
- Jenkinsfile structure
- Build → Test stages

Month 2 — Intermediate Jenkins + Automation (Weeks 5–8)

Week 5 — Git + Jenkins Integration

- Webhooks, Poll SCM
- Multibranch pipeline

Week 6 — Running Automation

- Run Maven/Gradle tests
- Cucumber tags
- Archive reports

Week 7 — Advanced Pipeline Skills

- Shared libraries
- Notifications
- Retry, timeout, post conditions

Week 8 — Docker + Jenkins

- Docker basics
- Jenkins agent in Docker

Month 3 — Framework Integration (Weeks 9–12)

Week 9 — JIRA + Xray Integration

- Xray REST API
- Upload test results
- Update status

Week 10 — Real CI/CD Project

- Full automation pipeline
- Versioned Jenkinsfile

Week 11 — Cloud CI/CD

- GitHub Actions
- GitLab CI
- Azure Pipelines

Week 12 — Portfolio Project

- Complete CI/CD setup
- End-to-end automation

Outcome After 3 Months

- ✓ Proficient Git & GitHub
- ✓ Pipeline development
- ✓ CI/CD automation
- ✓ Cucumber execution
- ✓ Xray integration
- ✓ Portfolio CI/CD project